BATTERY SAFETY

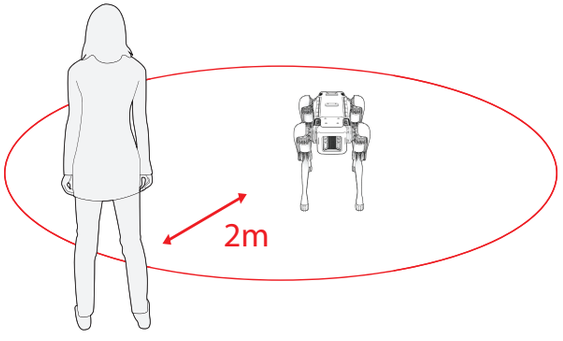
Spot is powered by a removable lithium ion battery pack. To reduce the risk of fire or electric shock:

- Only use the Spot battery, provided by Boston Dynamics.
- Only charge battery with provided Boston Dynamics charger.
- Don't short-circuit, burn, disassemble, submerge, puncture, crush, drop, or damage battery.
- If battery fire starts, don't try to put it out. Evacuate to a safe area and call fire department. Battery fires create toxic fumes and cannot be put out with conventional fire extinguishers or water.
- Store at -30 to 25C (-22 to 77F). Charge at 0 to 40C (32 to 104F). Operate at -20 to 45C (-4 to 113F).
- Spot operators should develop a battery storage and charging safety policy consistent with industry standards and local regulations.
- Remove battery from robot during transportation or storage.

BYSTANDER SAFETY

The operator is responsible for ensuring that all people who may come into close proximity of Spot are aware of robot hazards and receive Spot safety training.

- Keep at least 2m away from Spot and avoid the robot's path.
- It can be difficult to predict which way Spot will move next.
- Anyone passing Spot during autonomous patrol should avoid its path and maintain a 2m distance between Spot at all times.
- Do not stand between Spot and wall if this could violate the 2m or greater acceptable safety parameter.
- Never stand downhill or otherwise below Spot.

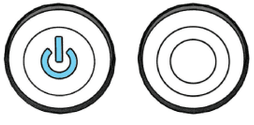


MOTOR LOCKOUT

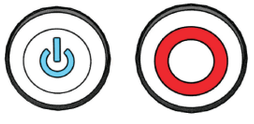
The lockout button mechanically disconnects the motor power and can only be activated by manually pressing the lockout button. Software cannot turn on the motors while the lockout is engaged.

To handle Spot safely, the robot must be placed in lockout mode. To enable lockout:

1. Turn motor power off with the controller.
2. Push the lockout button.
3. Confirm red light is off.



RED LIGHT OFF: Motor lockout engaged.



RED LIGHT ON: Motor not locked out; see all states for this button in **Lights** section on page 6.

It will be necessary to handle Spot when packing or unpacking for transport, replacing the battery, or righting Spot when it falls and is unable to self-right.

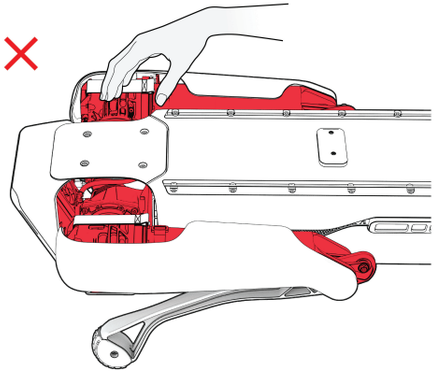
CAUTION: Never handle Spot unless motor lockout is engaged.

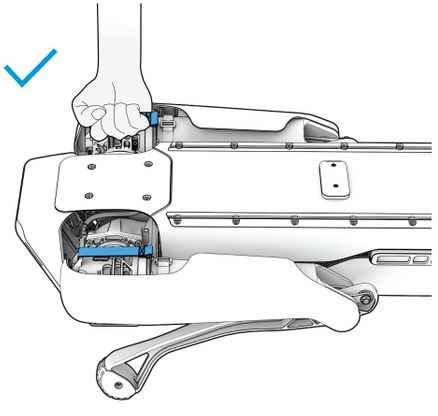
- Don't attempt to lift or move Spot while it is standing.
- Only move joints with motors in lockout mode or while Spot is powered off.
- Be careful to avoid pinch points (see page 12).

HANDLES

Spot has a handle at each hip joint. Use these to lift, carry, and roll the robot. When using the handles, always make a fist with each hand.

CAUTION: To avoid pinch points, never carry Spot with fingers extended.

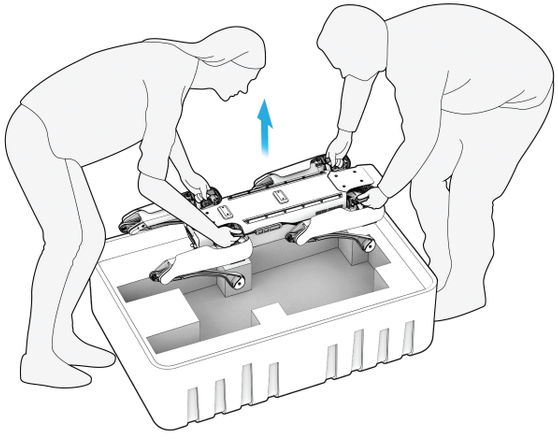




HANDLING SPOT

LIFTING SPOT

Spot is heavy (32 kg/70.5 lbs) and can be awkward to carry, particularly with attached payloads. Grab the handles and maintain safe lifting precautions. Always use two people to lift Spot, one at the front and one at the rear.



PLACING SPOT IN CASE

Remove battery from Spot. With two people, lift Spot by the handles and carefully lower into case.

